

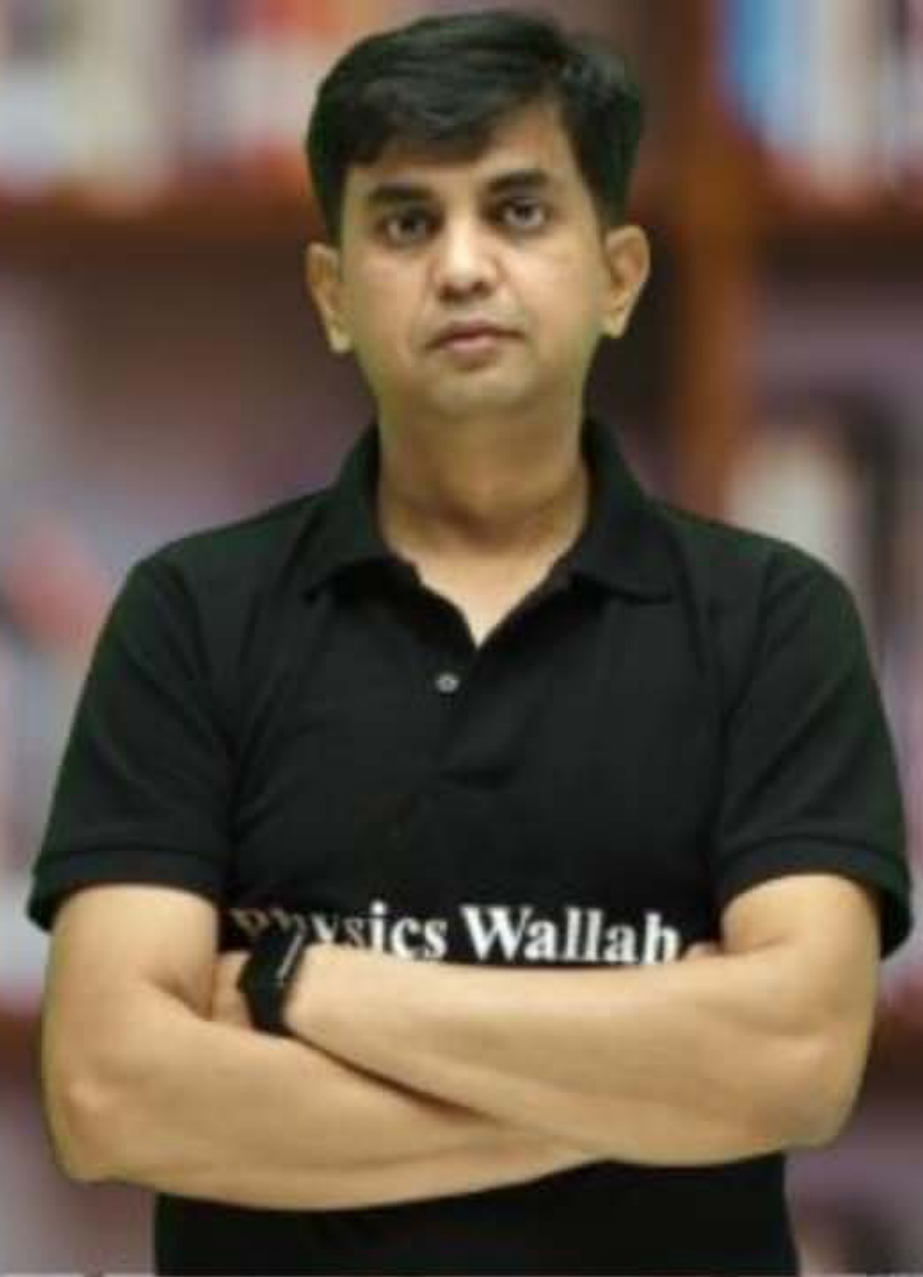
CS & IT ENGINEERING



Computer Network

IPv4 Header

Lecture No. - 03



By - Abhishek Sir



Recap of Previous Lecture



Topic

Identification Number

Topic

Fragmentation at Source Host



Topics to be Covered



Topic

Fragmentation at Source Host

Topic

Fragmentation at Router



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW





Topic : IPv4 Packet Header



0			16			31		
VER		HLEN		Type of Services		Total Length		
Identification Number				0DFM		Fragmentation Offset		
Time-to-Live		Protocol Type		Header Checksum				
Source IPv4 Address (32 - bits)								
Destination IPv4 Address (32 - bits)								
Optional Header (Options)								
Payload								

[NAT]

[GATE-2014] [2 Mark]



#Q. Every host in an IPv4 network has a (1-second resolution real-time clock with battery backup). Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around?

Ans: 262.144

IIT KGP Ans: 256

Design a 50-bit globally unique ID.

IPv4 Address : 32 bits

Identification Number : 18 bits

Total number of unique identifiers = 2^{18}
 [In each host, before wrap-around]

1000 identifiers → 1 Sec

2^{18} identifiers → $\frac{2^{18}}{10^3} \text{ sec} = 262.144 \text{ sec}$

[MCQ]

[GATE-2013] [2 Mark]

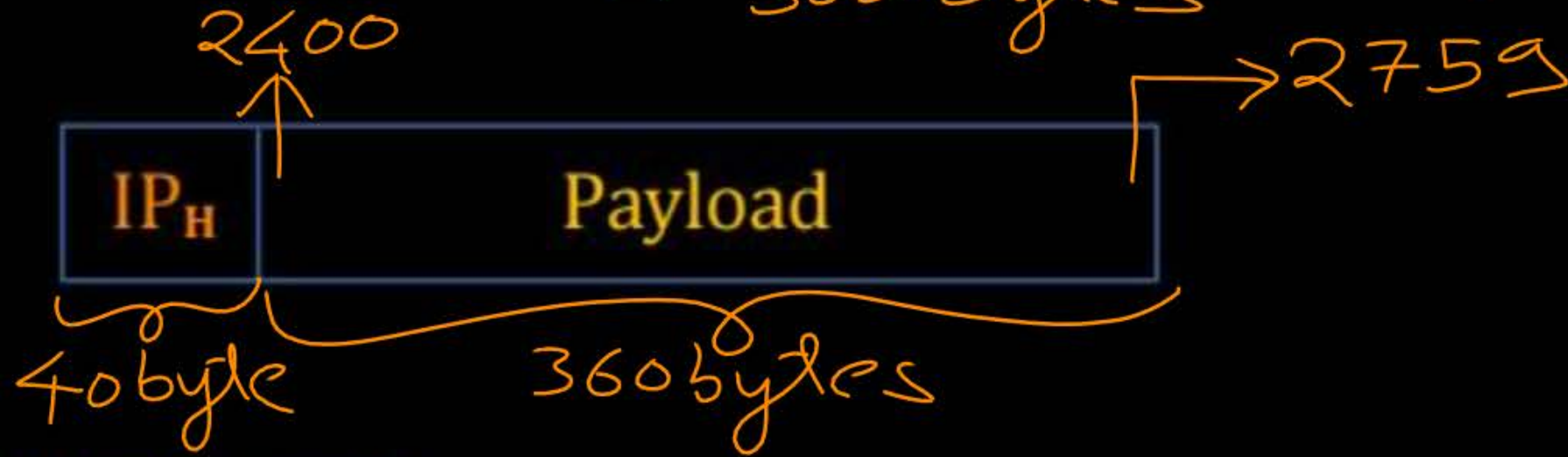


#Q. In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are

- ☒ A Last fragment, 2400 and 2789
- ☒ B First fragment, 2400 and 2759
- ☒ C Last fragment, 2400 and 2759
- ☒ D Middle fragment, 300 and 689

Ans: C

$$\begin{aligned} \text{Size of Payload} &= [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes} \\ &= [400 - (10 * 4)] \text{ bytes} \\ &= 360 \text{ bytes} \end{aligned}$$



$$\begin{aligned} \text{HLEN} &= 10 \\ \text{TL} &= 400 \\ \text{Offset} &= 300 \\ \text{MF bit} &= 0 \end{aligned}$$

Last fragment.

$$\begin{aligned} \text{First byte sequence number} &= \text{Offset} * 8 \\ &= 300 * 8 = 2400 \end{aligned}$$

$$\begin{aligned} \text{Last byte sequence number} &= (\text{Offset} * 8) + \text{Payload Size} - 1 \\ &= 2400 + 360 - 1 \\ &= 2759 \end{aligned}$$

[MCQ]

[GATE-2015] [2 Mark]



#Q. Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU = 1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?

- ☒ A 6 and 925
- ☒ B 6 and 7400
- ☒ C 7 and 1110
- ☐ D 7 and 8880

Ans: C

UDP Header Size = [8 bytes]
UDP Payload Size = [8880 bytes]

UDP Segment Size = 8888 bytes

MTU = [1500 bytes]
IPv4 Header Size = [20 bytes]

Maximum Payload Size = [MTU - Header Size] bytes
 = [1500 - 20] bytes
 = 1480 bytes

$$\text{UDP Segment Size} = 8888 \text{ bytes}$$

$$\text{Total Number of IP fragments } [N] = \left\lceil \frac{\text{UDP Segment Size}}{\text{Payload Size}} \right\rceil$$

$$N = \left\lceil \frac{8888 \text{ bytes}}{1480 \text{ bytes}} \right\rceil$$

$$= \lceil 6.005 \rceil$$

$$N = 7$$

$$\text{Offset value of the last IP fragment} = \left[\frac{(N-1) * \text{Payload Size}}{8} \right]$$

$$= \left[\frac{(7-1) * 1480}{8} \right]$$

$$= 1110$$

[MCQ]

[GATE-2025] [1 Mark]



#Q. Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?

- ☒ A 5 fragments, 1488 bytes
- ☐ B 6 fragments, 88 bytes
- ☒ C 6 fragments, 108 bytes
- ☐ D 6 fragments, 116 bytes

Ans: D

$$\begin{aligned}\text{UDP Header Size} &= [8 \text{ bytes}] \\ \text{UDP Payload Size} &= [7488 \text{ bytes}]\end{aligned}$$

$$\text{UDP Segment Size} = 7496 \text{ bytes}$$

$$\begin{aligned}\text{MTU} &= [1500 \text{ bytes}] \\ \text{IPv4 Header Size} &= 20 \text{ bytes}\end{aligned}$$

$$\begin{aligned}\text{Maximum Payload Size} &= [\text{MTU} - \text{Header Size}] \text{ bytes} \\ &= (1500 - 20) \text{ bytes} \\ &= 1480 \text{ bytes}\end{aligned}$$

UDP Segment Size = 7496 bytes

$$\text{Total Number of IP fragments [N]} = \left\lceil \frac{\text{UDP Segment Size}}{\text{Payload Size}} \right\rceil$$

$$N = \left\lceil \frac{7496 \text{ bytes}}{1480 \text{ bytes}} \right\rceil$$

$$= \lceil 5.06 \rceil$$

$$N = 6$$

Total length of the last IP fragment =

Header Size + [UDP Segment Size - (N-1) * Payload Size] bytes

$$= 20 \text{ byte} + [7496 \text{ bytes} - (6-1) * 1480 \text{ bytes}]$$

$$= (20 + 96) \text{ bytes}$$

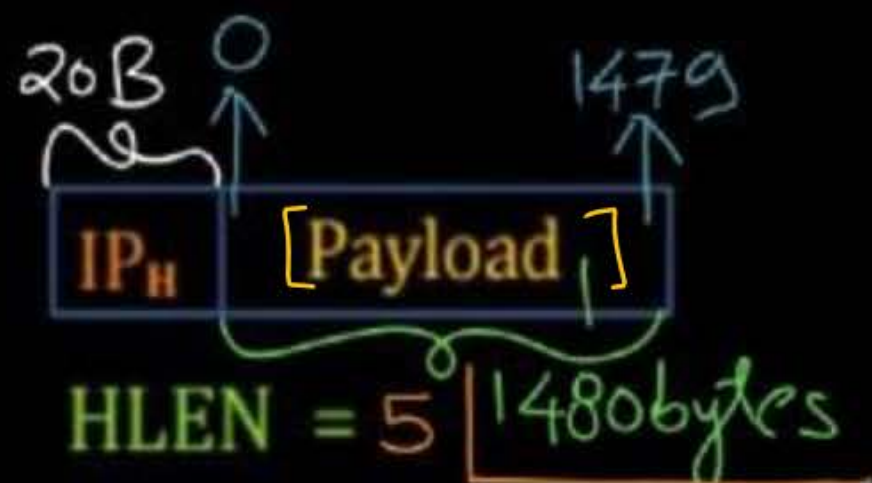
$$= 116 \text{ bytes}$$

Example 8 :-

[NAT]



#Q. Consider following IPv4 datagrams carrying one UDP segment (created in example 7) are arrived at intermediate IPv4 router. The next network (link) MTU at router is 500 bytes, calculate total number of IPv4 fragments required to carry the UDP segment after fragmentation?



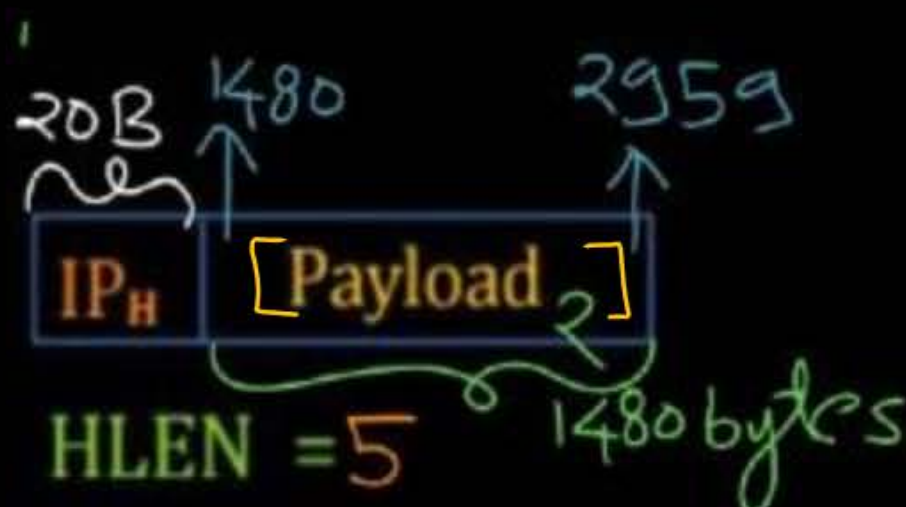
HLEN = 5 | 1480 bytes

TL = 1500

Id No. = Y

Offset = $\frac{0}{8} = 0$

MF bit = 1



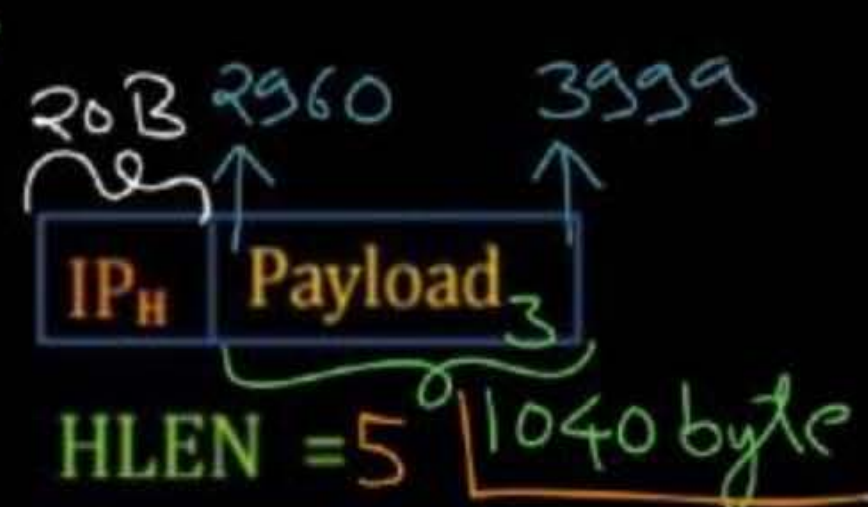
HLEN = 5 | 1480 bytes

TL = 1500

Id No. = Y

Offset = $\frac{1480}{8} = 185$

MF bit = 1



HLEN = 5 | 1040 byte

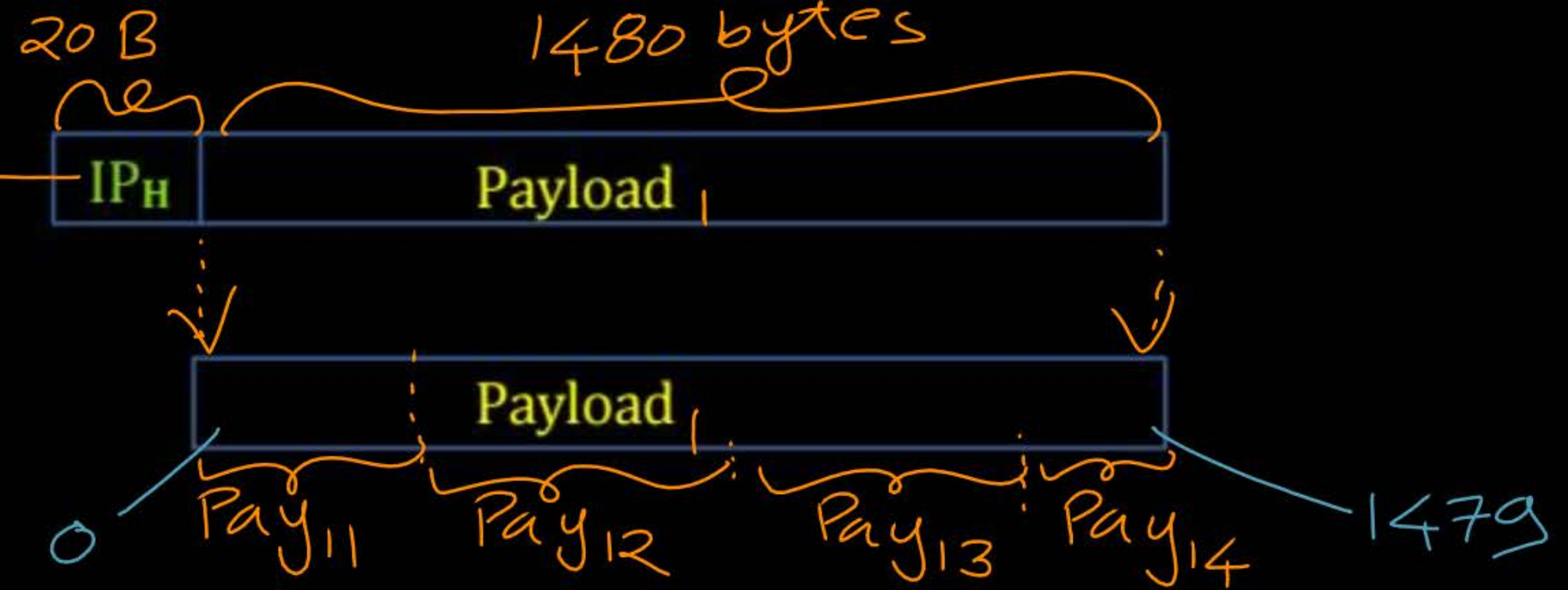
TL = 1060

Id No. = Y

Offset = $\frac{2960}{8} = 370$

MF bit = 0

HLEN = 5 ✓
TL = 1500
Id No. = Y
Offset = 0
MF bit = 1



20 B 480 byte
 IP_H Payload₁₁
 HLEN = 5
 TL = 500
 Id No. = Y
 Offset = $\frac{0}{8} = 0$
 MF bit = 1

20 B 480 Byte
 IP_H Payload₁₂
 HLEN = 5
 TL = 500
 Id No. = Y
 Offset = $\frac{480}{8} = 60$
 MF bit = 1

20 B 480 byte
 IP_H Payload₁₃
 HLEN = 5
 TL = 500
 Id No. = Y
 Offset = $\frac{960}{8} = 120$
 MF bit = 1

20 B 40 byte
 IP_H Payload₁₄
 HLEN = 5
 TL = 60
 Id No. = Y
 Offset = $\frac{1440}{8} = 180$
 MF bit = 1

$$\text{HLEN} = \underline{5}$$

$$\text{Total Length} = \underline{1500}$$

$$\text{Old Payload Size} = [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes}$$

↑

$$= [1500 - (5 * 4)] \text{ bytes}$$

$$= 1480 \text{ bytes}$$

$$\text{MTU} = \underline{500 \text{ bytes}}$$

$$\text{New Payload Size} = [\text{MTU} - (\text{HLEN} * 4)] \text{ bytes}$$

$$= [500 - (5 * 4)] \text{ bytes}$$

$$= 480 \text{ bytes}$$

$$\text{Total Number of IP fragments [N]} = \left\lceil \frac{\text{Old Payload Size}}{\text{New Payload Size}} \right\rceil$$

$$N = \left\lceil \frac{1480 \text{ bytes}}{480 \text{ bytes}} \right\rceil$$

$$= \lceil 3.08 \rceil$$

$$N = 4$$

Total length of the last IP fragment =

$$[\text{HLEN} * 4] + [\text{Old Payload Size} - (N-1) * \text{New Payload Size}] \text{ bytes}$$

$$= (5 * 4) + [1480 - (4-1) * 480] \text{ bytes}$$

$$= (20 + 40) \text{ bytes}$$

$$= 60 \text{ bytes}$$

$$\text{Offset value of the last IP fragment} = \text{Old offset value} + \left[\frac{(N-1) * \text{New Payload Size}}{8} \right]$$

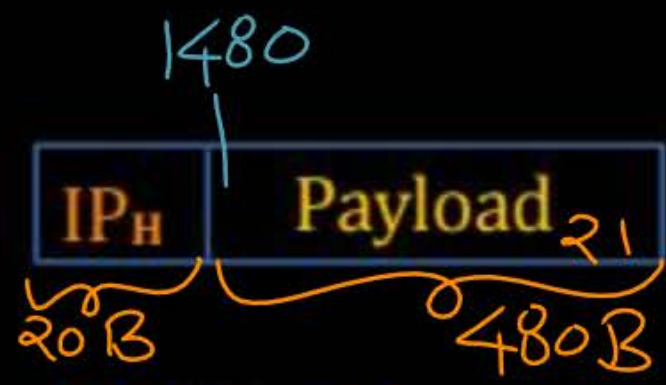
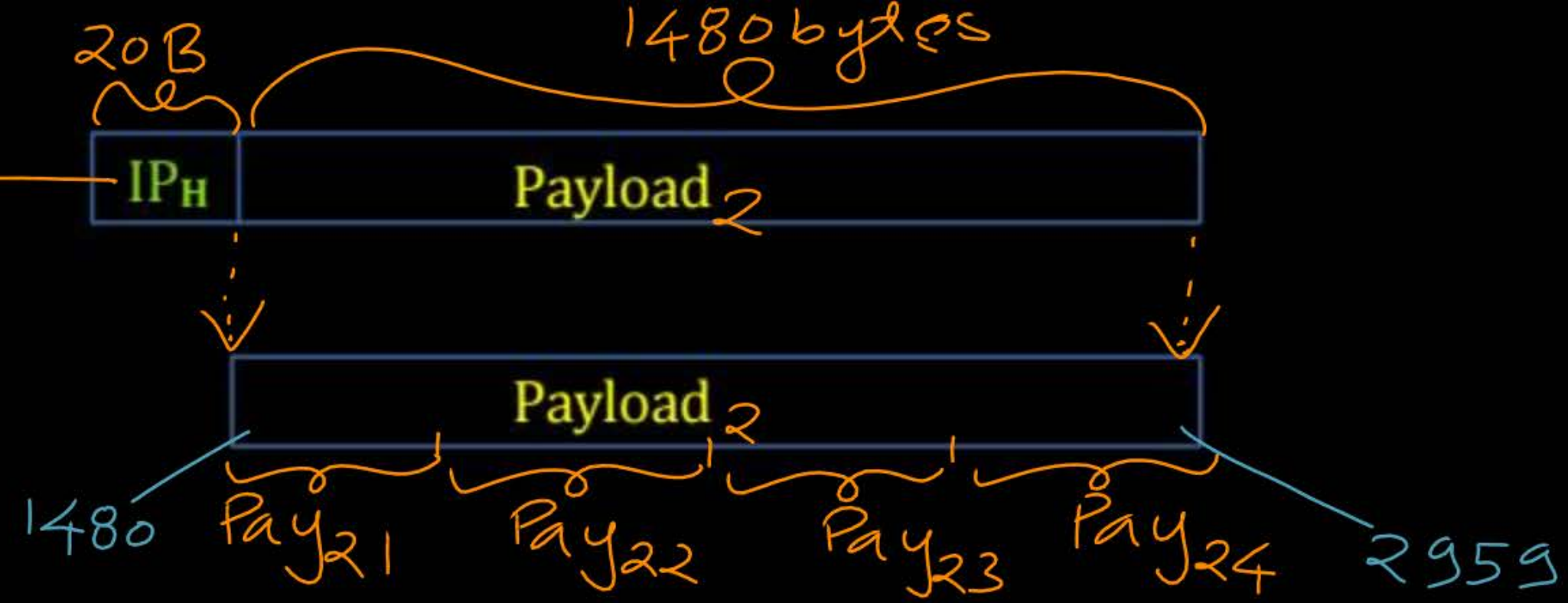
$$= 0 + \left[\frac{(4-1) * 480}{8} \right]$$

$$= 0 + \left[\frac{1440}{8} \right]$$

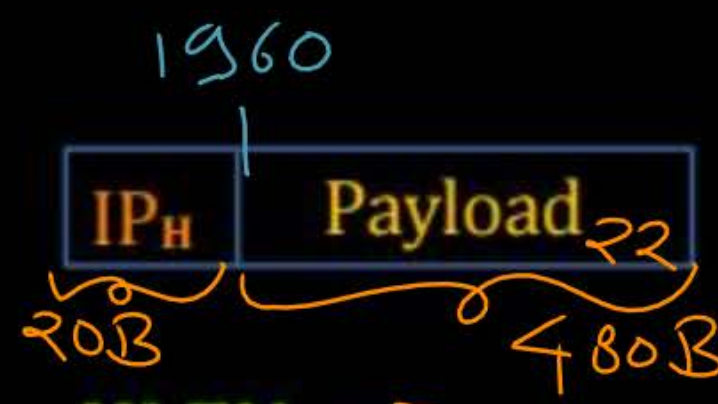
$$= 0 + 180$$

$$= 180$$

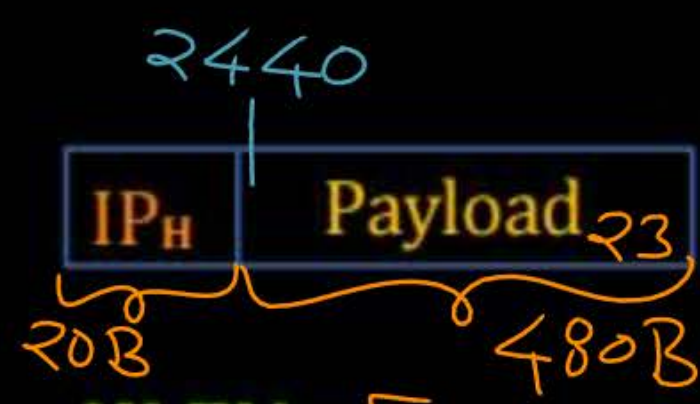
HLEN = 5
TL = 1500
Id No. = Y
Offset = 185
MF bit = 1



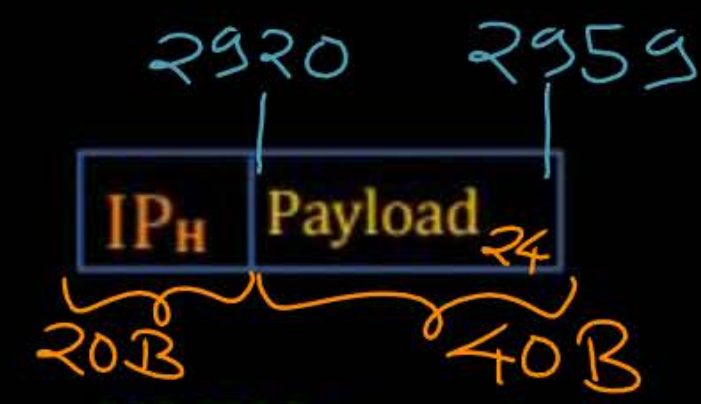
HLEN = 5
TL = 500
Id No. = Y
Offset = $\frac{1480}{8} = 185$
MF bit = 1



HLEN = 5
TL = 500
Id No. = Y
Offset = $\frac{1960}{8} = 245$
MF bit = 1



HLEN = 5
TL = 500
Id No. = Y
Offset = $\frac{2440}{8} = 305$
MF bit = 1



HLEN = 5
TL = 60
Id No. = Y
Offset = $\frac{2920}{8} = 365$
MF bit = 1

$$\text{HLEN} = \boxed{5}$$

$$\text{Total Length} = \underline{1500}$$

$$\begin{aligned} \text{Old Payload Size} &= [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes} \\ &= [1500 - (5 * 4)] \text{ bytes} \\ &= 1480 \text{ bytes} \end{aligned}$$

$$\underline{\text{MTU}} = \underline{500 \text{ bytes}}$$

$$\begin{aligned} \text{New Payload Size} &= [\text{MTU} - (\text{HLEN} * 4)] \text{ bytes} \\ &= [500 - (5 * 4)] \text{ bytes} \\ &= 480 \text{ bytes} \end{aligned}$$

$$\text{Total Number of IP fragments [N]} = \left\lceil \frac{\text{Old Payload Size}}{\text{New Payload Size}} \right\rceil$$

$$= \left\lceil \frac{1480 \text{ bytes}}{480 \text{ bytes}} \right\rceil$$

$$N = 4$$

Total length of the last IP fragment =

$$[\text{HLEN} * 4] + [\text{Old Payload Size} - (N-1) * \text{New Payload Size}] \text{ bytes}$$

$$= (5 * 4) + [1480 - (4-1) * 480] \text{ bytes}$$

$$= (20 + 40) \text{ bytes}$$

$$= 60 \text{ bytes}$$

$$\begin{aligned}
 \text{Offset value of the last IP fragment} &= \text{Old offset value} + \frac{(N-1) * \text{New Payload Size}}{8} \\
 &= 185 + \left[\frac{(4-1) * 480}{8} \right] \\
 &= 185 + \left[\frac{1440}{8} \right] \\
 &= 185 + 180 \\
 &= 365
 \end{aligned}$$



11th W.

HLEN = 5
TL = 1060
Id No. = Y
Offset = 370
MF bit = 0



HLEN =
TL =
Id No. = Y
Offset =
MF bit =



HLEN =
TL =
Id No. = Y
Offset =
MF bit =



HLEN =
TL =
Id No. = Y
Offset =
MF bit =



$$\text{HLEN} = 5$$

$$\text{Total Length} = 1060$$

$$\text{Old Payload Size} = [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes}$$

$$\text{MTU} = 500 \text{ bytes}$$

$$\text{New Payload Size} = [\text{MTU} - (\text{HLEN} * 4)] \text{ bytes}$$

$$\text{Total Number of IP fragments [N]} = \frac{\text{Old Payload Size}}{\text{New Payload Size}}$$

Total length of the last IP fragment =

$$[\text{HLEN} * 4] + [\text{Old Payload Size} - (N-1) * \text{New Payload Size}] \text{ bytes}$$

$$\text{Offset value of the last IP fragment} = \text{Old offset value} + \frac{(N-1) * \text{New Payload Size}}{8}$$



Topic : Fragmentation at Intermediate Router

$$\text{IPv4 Datagram Size} \leq \text{Intermediate Network MTU}$$

$$\text{Old Payload Size} = [\text{Total Length} - (\text{HLEN} * 4)] \text{ bytes}$$

$$\text{New Payload Size} = [\text{MTU} - (\text{HLEN} * 4)] \text{ bytes}$$

$$\text{Total Number of IP fragments [N]} = \left\lceil \left[\text{Old Payload Size} / \text{New Payload Size} \right] \right\rceil$$

$$\begin{aligned} \text{Total length of the last IP fragment} &= \\ &[\text{HLEN} * 4] + [\text{Old Payload Size} - (\text{N}-1) * \text{New Payload Size}] \text{ bytes} \end{aligned}$$

$$\begin{aligned} \text{Offset value of the last IP fragment} &= \\ &\text{Old offset value} + [(\text{N}-1) * \text{New Payload Size} / 8] \end{aligned}$$

[NAT]

H.W.

ISSC

[GATE-2016] [2 Mark]



#Q. An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is _____.

[MCQ]

H.W

IIT-KGP **[GATE-2014] [2 Mark]**



#Q. An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are

- A** MF bit: 0, Datagram Length: 1444; Offset: 370
- B** MF bit: 1, Datagram Length: 1424; Offset: 185
- C** MF bit: 1, Datagram Length: 1500; Offset: 37
- D** MF bit: 0, Datagram Length: 1424; Offset: 2960

[MCQ]

H.W.

IIT-D

[GATE-CS-2004] [2 Mark]



#Q. Consider three IP networks A, B and C. Host H_A in network A sends messages each containing 180 bytes of application data to a host H_C in network C. The TCP layer prefixes a 20 byte header to the message. This passes through an intermediate network B. The maximum packet size, including 20 byte IP header, in each network is :

A : 1000 bytes

B : 100 bytes

C : 1000 bytes

Assuming that the packets are correctly delivered, how many bytes, including headers, are delivered to the IP layer at the destination for one application message, in the best case ? [Consider only data packets.]

A

200

B

220

C

240

D

260

[MCQ]

/ H.W.

IIT ->

[GATE-IT-2004] [2 Mark]



#Q. A TCP message consisting of 2100 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1200 bytes per datagram and the second network can carry a maximum payload of 400 bytes per datagram, excluding network overhead. Assume that IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission?

- ☐ A 40 bytes
- ☐ B 80 bytes
- ☐ C 120 bytes
- ☐ D 160 bytes



2 mins Summary



Topic

Fragmentation at Source Host

Topic

Fragmentation at Router



THANK - YOU

